

# ENERGY DATA VISUALISATION HACKATHON

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## Context & Background Documents

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|--------------|-----------------------------------------------------------------|
| Version      | 1.0 — May 2026                                                  |
| Audience     | Hackathon participants                                          |
| Purpose      | Reference material for all 3 challenge axes                     |
| Axes covered | Electrification · Load Shifting & Myth Busting · Open Challenge |

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## 1. Belgian Electricity Bill Explainer

What makes a quality explainer? Plain language, concrete examples, a visual structure, and no jargon without definition. Every concept must answer: 'what does this mean for my bill?'

### What is your electricity bill made of?

Your electricity bill in Belgium is not a single price — it is the sum of three distinct components, each regulated and billed differently. Understanding this structure is the foundation for any visualisation or cost comparison built during this hackathon.

|                 |                                                                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component       | Description                                                                                                                                                                                                            |
| Taxes & Excises | Fixed levies set by federal and regional governments. Include VAT (21%), excise duties, and contributions to renewable energy support schemes. Non-negotiable, same for all suppliers.                                 |
| Grid Fees       | Charged by the distribution system operator (DSO) for use of the electricity network. In Brussels: Sibelga. Fixed annually, independent of your energy supplier. Cover infrastructure maintenance and public lighting. |
| Commodity Price | The cost of the electricity itself. Set by your chosen supplier. Since 2007, Belgian households can freely choose their supplier. Can be                                                                               |

fixed, bi-hourly, or dynamic (hourly).

## Fixed vs. Dynamic Tariffs

The commodity price is the only component that can vary — and it is the lever that makes demand flexibility possible.

- **Fixed tariff:** One price per kWh, stable over a contract period (typically 1–3 years). Predictable but does not reflect real-time grid conditions.
- **Bi-hourly tariff (day/night):** Two price levels — cheaper at night and weekends. The traditional way to incentivise off-peak consumption.
- **Dynamic (hourly) tariff:** Price changes every hour based on wholesale market prices. The day is split into time zones. Rewards flexibility — but requires awareness or automation (HEMS).

*Key insight for the hackathon: most households on a dynamic tariff never see the hourly price signal. The visualisation gap is the opportunity.*

## Smart Meters & Data Granularity

Smart meters in Belgium capture consumption data at different granularities depending on their type:

- **R1 meters:** Data captured hourly, transmitted in 5 daily time zones. Standard for residential customers.
- **R3 meters:** Data captured every 15 minutes. More granular — used for commercial customers and advanced analysis.

This data is transmitted by the DSO (Sibelga in Brussels, Fluvius in Flanders) to energy suppliers, who use it to calculate the commodity portion of the bill.

*For hackathon participants: the EnergieID dataset provides anonymised load curves at household level — use it as your primary source for consumption pattern analysis.*

## The 2027 Sibelga Capacity Tariff

From 2027, Sibelga will introduce a capacity-based grid tariff for Brussels households. Instead of paying purely per kWh consumed, households will be charged based on their peak power demand (measured in kVA).

- **What this means:** A household that consumes the same monthly kWh but concentrates usage during evening peaks will pay more than one that spreads consumption evenly.
- **The ranges:** 0–13 kVA and 13–56 kVA. Crossing a threshold triggers a higher grid fee.
- **The implication:** Shifting a washing machine from 18:00 to 23:00 is no longer just about the commodity price — it can directly reduce your capacity tariff tier.

*This tariff model is a core dataset for Challenge 1 (Electrification) and Challenge 2 (Load Shifting). Use it to make cost comparisons concrete and forward-looking.*

## 2. Sibelga Capacity Tariff Explainer (2027)

*What makes a quality technical explainer? It must translate regulatory complexity into household-level impact. Every abstract concept needs a concrete example with real numbers.*

### What is a capacity tariff?

A capacity tariff (also called a peak tariff) is a grid fee structure where you are charged based on the maximum power you draw from the grid at any given moment — not just the total energy you consume over a month.

This model is being progressively introduced across Belgium to incentivise households to flatten their consumption curve and avoid simultaneous peak loads that stress the grid.

### How it works in Brussels (from 2027)

- **Measurement unit:** kVA (= kilovolt-ampere) — a measure of instantaneous power demand, not energy consumed over time.
- **Billing logic:** Sibelga measures your peak power demand each month. Your grid fee is set according to which capacity bracket you fall into.
- **Brackets:** 0–13 kVA (standard residential) and 13–56 kVA (higher consumption — e.g. households with EV chargers, heat pumps, and heavy appliances running simultaneously).
- **No weekend distinction:** Unlike the old bi-hourly tariff, the capacity tariff applies 7 days a week.

### A concrete example

Two households consume exactly 300 kWh in June.

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Household A | Runs dishwasher, washing machine, and EV charger simultaneously at 19:00. Peak demand: 14 kVA → upper bracket → higher grid fee. |
| Household B | Same appliances, staggered across the day. Peak demand: 8 kVA → lower bracket → lower grid fee.                                  |
| Difference  | Same total consumption. Same commodity cost. Different grid fee — purely because of when they consume.                           |

*This example is directly usable in Challenge 1 and 2 visualisations. Show the financial impact of load shifting in euros, not just in abstract kWh or kVA.*

### Why it matters for this hackathon

- It makes load shifting financially tangible for households — not just a civic gesture but a money-saving behaviour.
- It creates a new dimension for cost comparison: the same appliance costs differently depending on when it runs.
- It is forward-looking: teams that factor in the 2027 tariff build tools with real longevity.

## 3. Glossary of Key Terms

*What makes a quality glossary? Definitions must be plain-language, context-specific, and ordered for progressive understanding — not alphabetical. Each term should answer: 'why does this matter for my project?'*

### Grid & Network

|                                           |                                                                                                                                                                                                        |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TSO — Transmission System Operator</b> | Manages the high-voltage electricity backbone (400 kV to 30 kV). In Belgium: Elia. Responsible for real-time balance between national production and consumption.                                      |
| <b>DSO — Distribution System Operator</b> | Manages the local electricity distribution network (low and medium voltage). Delivers power to households. In Brussels: Sibelga. In Flanders: Fluvius.                                                 |
| <b>Load curve</b>                         | A graph showing a household's or network's electricity consumption over time (usually 15-min or hourly intervals). The foundational dataset for demand analysis.                                       |
| <b>Peak demand</b>                        | The highest level of electricity consumption recorded in a given period. Evening peaks (17:00–21:00) are Belgium's biggest grid challenge.                                                             |
| <b>Grid congestion</b>                    | When electricity demand exceeds network capacity in a specific area. Leads to higher costs and infrastructure investment needs.                                                                        |
| <b>Inertia</b>                            | The physical resistance of large rotating generators (nuclear, gas) to sudden frequency changes. Renewables like solar PV have no inertia — making real-time grid balance harder as their share grows. |

### Tariffs & Billing

|                              |                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>kWh — kilowatt-hour</b>   | The standard unit for measuring energy consumption. Running a 2,000W washing machine for 1 hour consumes 2 kWh.             |
| <b>kVA — kilovolt-ampere</b> | A unit of instantaneous power demand. Used in the 2027 Sibelga capacity tariff to measure peak load.                        |
| <b>Capacity tariff</b>       | A grid fee based on peak power demand (kVA), not just total consumption. Incentivises spreading consumption across the day. |
| <b>Dynamic tariff</b>        | An electricity supply contract where the commodity price varies by the hour, reflecting real-time wholesale market prices.  |

|                         |                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bi-hourly tariff</b> | The traditional 'day/night' tariff — cheaper electricity at night and weekends. Being phased out in favour of more granular pricing.                                           |
| <b>Capacity factor</b>  | The ratio of actual energy produced by a power source vs. its theoretical maximum. Nuclear: ~95%. Solar PV: ~35%. Wind: ~25–40%. Critical for comparing energy sources fairly. |

## Technology & Systems

|                                             |                                                                                                                                                                                                                                              |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HEMS — Home Energy Management System</b> | A system that automatically manages household energy flows — deciding when to charge a battery, run a washing machine, or export solar power. Examples: Home Assistant (open-source), Schneider Electric. Requires smart meter connectivity. |
| <b>Smart meter (R1 / R3)</b>                | R1: captures data hourly, transmitted in 5 daily time zones. R3: captures every 15 minutes. Both transmit load curves (consumption and injection) to the DSO.                                                                                |
| <b>PV — PhotoVoltaire</b>                   | Solar panels that convert sunlight into electricity. Produce energy during daylight hours — peaking at noon — creating a mismatch with evening household demand peaks.                                                                       |
| <b>Load shifting</b>                        | The act of moving electricity consumption from peak hours to off-peak hours. E.g. running a dishwasher at 23:00 instead of 19:00. The primary lever for flattening the consumption curve.                                                    |
| <b>Peak shaving</b>                         | Reducing the height of consumption peaks, either by shifting load or using storage (batteries). Reduces grid stress and, under capacity tariffs, lowers household bills.                                                                     |
| <b>Injection</b>                            | Sending excess electricity (e.g. from solar panels) back into the grid. Metered separately from consumption. Grid fees apply regardless of injection volume.                                                                                 |

## Organisations & Programmes

|                                           |                                                                                                                                                                                              |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EC — Energy Community</b>              | A local cooperative model where members collectively produce, share, and manage energy. Members benefit from prices set between the wholesale injection rate and the standard supplier rate. |
| <b>EUSEW — EU Sustainable Energy Week</b> | Europe's largest annual event dedicated to renewables and energy efficiency. Takes place 9–11 June 2026 in Brussels — this hackathon runs immediately after.                                 |
| <b>EnergieID</b>                          | A Belgian cooperative platform for monitoring and verifying energy, water and waste consumption. Provides the primary open dataset for this hackathon (10,000 households).                   |
| <b>Sibelga</b>                            | The DSO (= Distribution System Operator) for the Brussels Capital Region. Manages the electricity and gas distribution network for all 19 Brussels municipalities.                           |
| <b>Elia</b>                               | Belgium's TSO (= Transmission System Operator). Manages the high-voltage grid and publishes real-time production, consumption, and renewable generation data via open data portal.           |
| <b>Fluvius</b>                            | The DSO for Flanders. Manages smart meter rollout (mandatory since 2023) and publishes granular consumption datasets.                                                                        |

## 4. Brussels Energy Transition — Context & Background

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*What makes a quality context document? It must frame the problem without lecturing. Facts over opinions. Local relevance over generic EU narratives. Actionable framing — every section should connect to a hackathon challenge.*

### Where Belgium stands today

Belgium's electricity mix is in transition. As of 2024, nuclear energy accounts for approximately 45–50% of electricity production, making it the dominant source. Renewables (wind, solar, biomass) represent around 20–25%, with the remainder coming from gas and imports.

Belgium is a net electricity importer at peak demand times — particularly winter evenings when renewable production is low and consumption is high. This structural imbalance is the core problem this hackathon addresses.

- **Nuclear:** High capacity factor (~95%), provides stable baseload. Belgium's nuclear exit has been delayed — reactors Doel 4 and Tihange 3 will now operate until at least 2035.
- **Solar PV:** Growing rapidly on rooftops across Brussels. Capacity factor ~35% — produces at noon, not in the evening when demand peaks.
- **Wind:** Offshore wind in the North Sea is Belgium's fastest-growing renewable. Still insufficient to cover peak demand alone.
- **Gas:** Belgium's federal and Elia strategy still includes gas as a backup ('centrale au gaz'). Creates coherence challenges with decarbonisation goals.

### The Brussels household reality

Brussels is a dense, urban region of approximately 1.2 million inhabitants across 19 municipalities. Key characteristics relevant to energy transition:

- High proportion of renters (over 60%) — limiting ability to install solar panels or upgrade heating systems.
- Older housing stock — many buildings are poorly insulated, increasing heating demand.
- Gas heating dominates — most Brussels households still heat with gas boilers. Heat pump adoption is low but growing.
- Limited EV (= Electric Vehicle) penetration — but growing rapidly, creating new evening peak pressure as charging concentrates between 18:00 and 22:00.
- Energy poverty is a real concern — approximately 15–20% of Brussels households experience difficulty paying energy bills.

*These realities define your target audience: a typical Brussels household is a renter, heating with gas, hesitant about electrification, and not yet equipped with a smart energy management system.*

## The daily production-consumption mismatch

This is the structural challenge at the heart of Challenges 2 and 3:

|                                             |                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Solar production peak</b>                | 12:00–14:00 — maximum solar generation. Grid is often in surplus. Electricity prices are lowest on dynamic tariffs. |
| <b>Household consumption trough</b>         | 12:00–16:00 — most people are at work. Low residential demand. Surplus energy is exported or curtailed.             |
| <b>Evening consumption peak</b>             | 17:00–21:00 — cooking, EV charging, heating all converge. Grid is under maximum stress. Prices are highest.         |
| <b>Renewable production at evening peak</b> | Near zero for solar. Wind is variable. Gas and nuclear must compensate. Imports increase.                           |

Load shifting — moving consumption from the evening peak to the midday trough — is one of the highest-impact, lowest-cost interventions available. It requires no new infrastructure. It requires awareness and the right tools.

## Why electrification is stalling — and what data can do

A large share of Brussels citizens plan to stay with gas heating and combustion engine cars. The barriers are not primarily financial — they are cognitive and informational:

- **Opaque costs:** The true 10-year cost of a heat pump vs. a gas boiler is not obvious. Upfront cost dominates perception; running cost savings are invisible.
- **Misinformation:** Common myths (see Challenge 2) actively discourage electrification. Many citizens believe renewables are more expensive, or that injecting solar energy cancels out their grid fees.
- **Complexity:** Dynamic tariffs, capacity tariffs, HEMS, smart meters — the system is genuinely hard to navigate without support.
- **Distrust:** Energy bills have increased sharply since 2021. Citizens are sceptical of promises about cheaper alternatives.

This is where data visualisation has a unique role. A clear, honest, evidence-based comparison can cut through complexity and distrust in ways that policy documents and marketing cannot. That is the mission of this hackathon.

*'The missing piece is not a subsidy — it's a clear, honest, visual comparison.'* — Challenge 1 framing

## Key data sources for context

- **Elia open data:** Real-time and historical production, consumption, import/export. [opendata.elia.be](https://opendata.elia.be)
- **EnergieID:** 10,000-household anonymised consumption dataset. Primary source for load curve analysis.
- **Sibelga:** Brussels grid fee structure and 2027 capacity tariff model.

- **IBSA Brussels:** Demographic and socioeconomic data for the Brussels Capital Region. [ibsa.brussels/opendata](https://ibsa.brussels/opendata)
- **IRM / Open-Meteo:** Historical weather data — essential for correlating solar production with consumption patterns. [opendata.meteo.be](https://opendata.meteo.be) / [open-meteo.com](https://open-meteo.com)